1. C1 toy ladder: standard base vs twisted all-lift line, n = 4

2026-10-27. Protocol `PROTOCOL.md` (frozen before the run). Raw rows and summary in `reports/run-20261027T1502Z/`. All numbers are exact counts; no wall-clock claim.

1.1. Question

The upstream C2PNB cost model assumes the per-decomposition solver cost c_{trial} is unchanged when the Gaudry-Diem base $x \in F_Q$ is replaced by a twisted line uF_Q ($T(u) = T\left(\frac{\sqrt{b}}{u}\right) = 0$), which makes every abscissa lift and raises the relations per trial by about $(2\rho)^4 \approx 16$. This is the toy ladder: E over F_{Q^4}, $Q = 2^k$, $k \in \{5, 7, 9, 11\}$ for yield and $k \in \{5, \ldots, 15\}$ for cost.

1.2. Method

- Yield: exact ordered counts of $R = P_1 + P_2 + P_3 + P_4$ by a pair table, same targets for both bases, empirical-variance z tests, paired bootstrap.
- Cost: S_5 by resultants, symmetrised; in characteristic 2 it is a polynomial in e_1^2, e_2^2, e_3^2, e_4. With $x_i = uy_i$: four equations over F_Q in $(E_1..E_4)$, weights $(2, 2, 2, 1)$, 105 monomials, weighted Bezout 512. Solver: weighted Macaulay matrix at $D = 26$, forward elimination with sparsest pivots; counted: F_Q multiplications and inversions.
- Curves: family A ($a, b \in F_{16}$, the task's), A0 ($\tau = 0$ control), NEG ($\mathrm{Tr}(a) = 1$), family B (generic b, u by search in $\ker T$).

1.3. Results

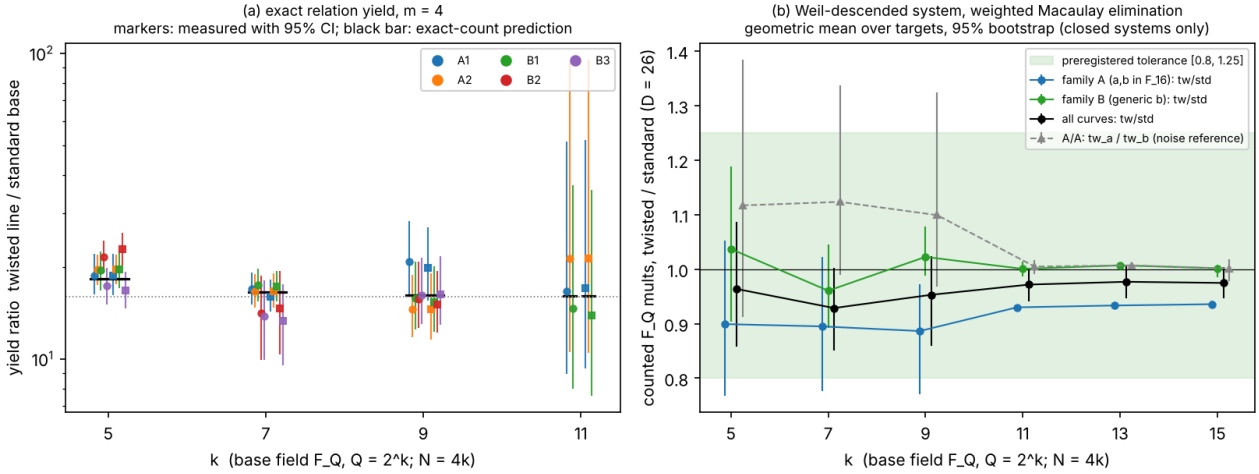


Figure 1: (a) yield ratio, measured with 95% CI and exact-count prediction. (b) cost ratio, geometric mean with 95% bootstrap, tolerance [0.8, 1.25].

Yield. H-Y2 passes: prediction inside the interval in 34/36, pooled observed 17.3254 against predicted 16.9227 (1.0238). H-Y4 passes exactly: with $\mathrm{Tr}(a) = 1$ the twisted line has 0 lifting abscissae and 0 relations at $k = 5, 7, 9$. H-Y5 (A/A) passes 18/21, but on A2 the two valid u give identical counts. H-Y1 fails (48/54 rows within $|z| \leq 3$; five rows beyond 5, all on A0: $k = 9$, $z = -5.17$, and four rows whose counts are all zero against a prediction near 15.5, kept in the gate with $z = -\infty$) and H-Y3 fails: A0 ($b = 1$) has extremely clustered counts, so the control is uninformative. 16678 relations enumerated at $k = 5$ all satisfy the descended equations (0 violations).

Cost. Matrix dimensions are identical on all 594 solves (4840 x 4200, 40.7 MB), rank and Hilbert function agree on 396/396 pairs, HF $= 512$ everywhere, first reached at $D = 25$. Counted mults ratio twisted/standard, both closed ($n = 384$): geometric mean **0.9606**, 95% [0.9279, 0.9913], inside the preregistered tolerance [0.8, 1.25]: **unchanged** for this solver. Family A 0.9097 [0.8627, 0.9539], family B 1.0045 [0.9788, 1.0327]; A/A noise reference 1.0596 [1.0043, 1.1266]. At $k \geq 11$ the spread vanishes (A about 0.93, B about 1.00). Per-solve ratios range 0.476 to 2.026 (87% inside the tolerance). 11 of 594 systems (1 standard, 10 twisted) do not close at $D = 26$: their top-weighted forms have a common zero at infinity (target-dependent, also on the standard base); a full F4 would be needed.

1.4. Limits

Simplified solver, not F4; no rational-solution extraction; no timing; $a, b \in F_{16}$ forces a τ structure (only two $\tau = 1$ classes), hence the extension family B; family B at $k \geq 9$ has no absolute yield prediction. The measured per-trial

elimination cost supports the model's "c_{trial} unchanged" assumption to within about 4% (generic) to 9% (family A) for this solver.